

Alexandro Gadja

Applied Electronics Student (UTCN, final year) | Electrical Installations

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Available for part-time / working student roles · ~20 h/week · Cluj-Napoca, Oradea or remote

PROFILE

Final-year Applied Electronics student at UTCN (graduating 2027) who designs on paper, verifies in simulation and builds in the field. I co-run a small family electrical installation business and completed a 200-hour internship at EnergoBit. Projects span FPGA design in VHDL, analog signal conditioning and PWM generation with full PSpice analysis. Native Romanian and Italian, fluent English.

EXPERIENCE

Co-owner · Electrical Installations — Aug 2025 – Present
Gadja Construct (family business) · Cluj-Napoca

- Prepare material orders and labor price quotes for the company's installation jobs; work on site on electrical installations.

Engineering Intern · Relay Protection & Design Department — Jul 2026 – Sep 2026 · 200 h
EnergoBit · Cluj-Napoca

- Supported the protection & design team with project documentation; attended in-house technical sessions on relay protection and electrical/electronic design.

Electrician's Assistant (internship) — Aug 2024 – Sep 2024
Elux Electrics · Cluj-Napoca

- Assisted electricians on site with installation work. Also built the company's website (eluxelectrics.ro).

Video Editor (Freelance, remote) — Apr 2022 – Sep 2022
US-based company

- Edited wedding films in Adobe Premiere Pro and After Effects to fixed deadlines for an English-speaking client.

PROJECTS

Digital Watch with Stopwatch & Time Zones on FPGA — VHDL · Vivado · Nexys A7-100T

- Finite-state machine with time setting (blinking digits), stopwatch mode and 3 switch-selectable time zones; button debouncing, clock dividers from 100 MHz and multiplexed 7-segment display driver.

4-Stage Analog Signal Conditioning Chain — ADA4077 · ±15 V

- Designed and dimensioned a 2-op-amp instrumentation amplifier, Sallen-Key band-pass filter, programmable-gain amplifier (10–16 dB in 2 dB steps) and precision full-wave rectifier.

PWM Signal Generator — OrCAD PSpice · TL081

- 12 kHz generator with 20–90% adjustable duty cycle and 3–10 V amplitude; validated with transient, DC sweep, parametric and Monte Carlo (component tolerance) analyses.

Room Thermostat — analog & digital front-end + LCD — LM35 · MCP6022 · ADC0808 · 80C51 · Proteus

- Selected the LM35 after comparing 5 sensor families; designed a rail-to-rail op-amp offset/gain stage (gain 14.3) mapping 5–40 °C to the 0–5 V range of an ADC0808; digital alternative with an MCP9801 over I2C on an 80C51, output on a 16x2 LCD. Interactive documentation: gadja4.github.io/uC3.1.

Languages: Romanian (native) · Italian (native / bilingual) · English (advanced, C1) · French, German (basic)
Driving licence: Category B